

Building Robotic Agents that Think and Act Like Humans

Jingyun Yang

Agenda

- **Part 1:** How can robotic agents discover and represent the hierarchical structure of events or tasks in our everyday lives?

- **Part 2:** How can robotic agents organize and compose simple skills and behaviors to complete complex tasks?

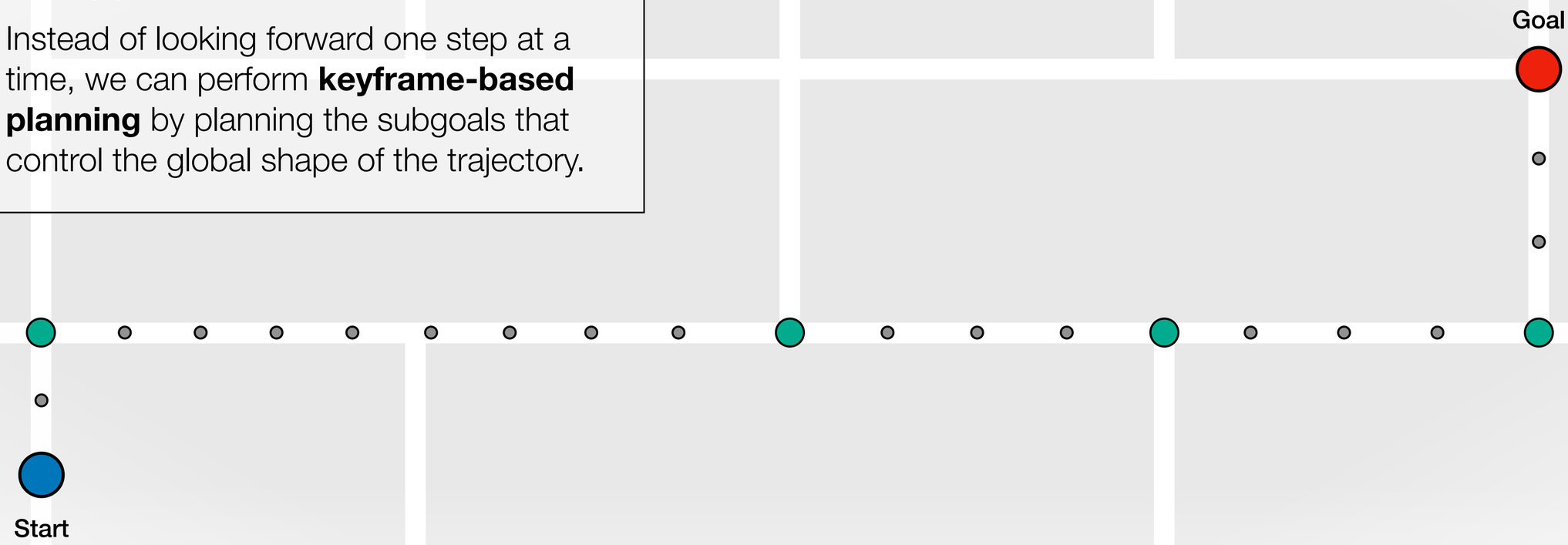
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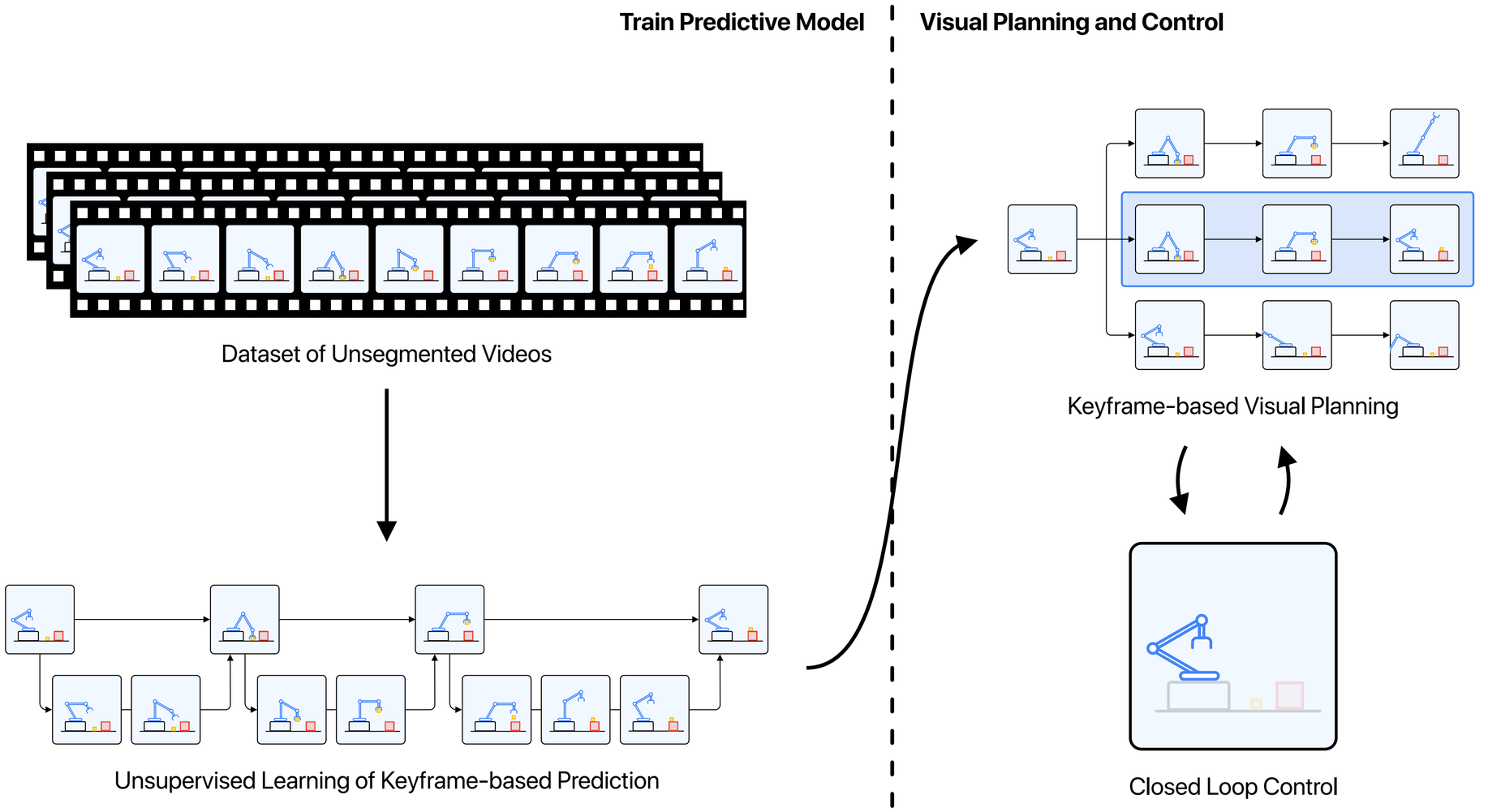
Consider making plans towards a far-away goal:

KEY INSIGHT

Instead of looking forward one step at a time, we can perform **keyframe-based planning** by planning the subgoals that control the global shape of the trajectory.

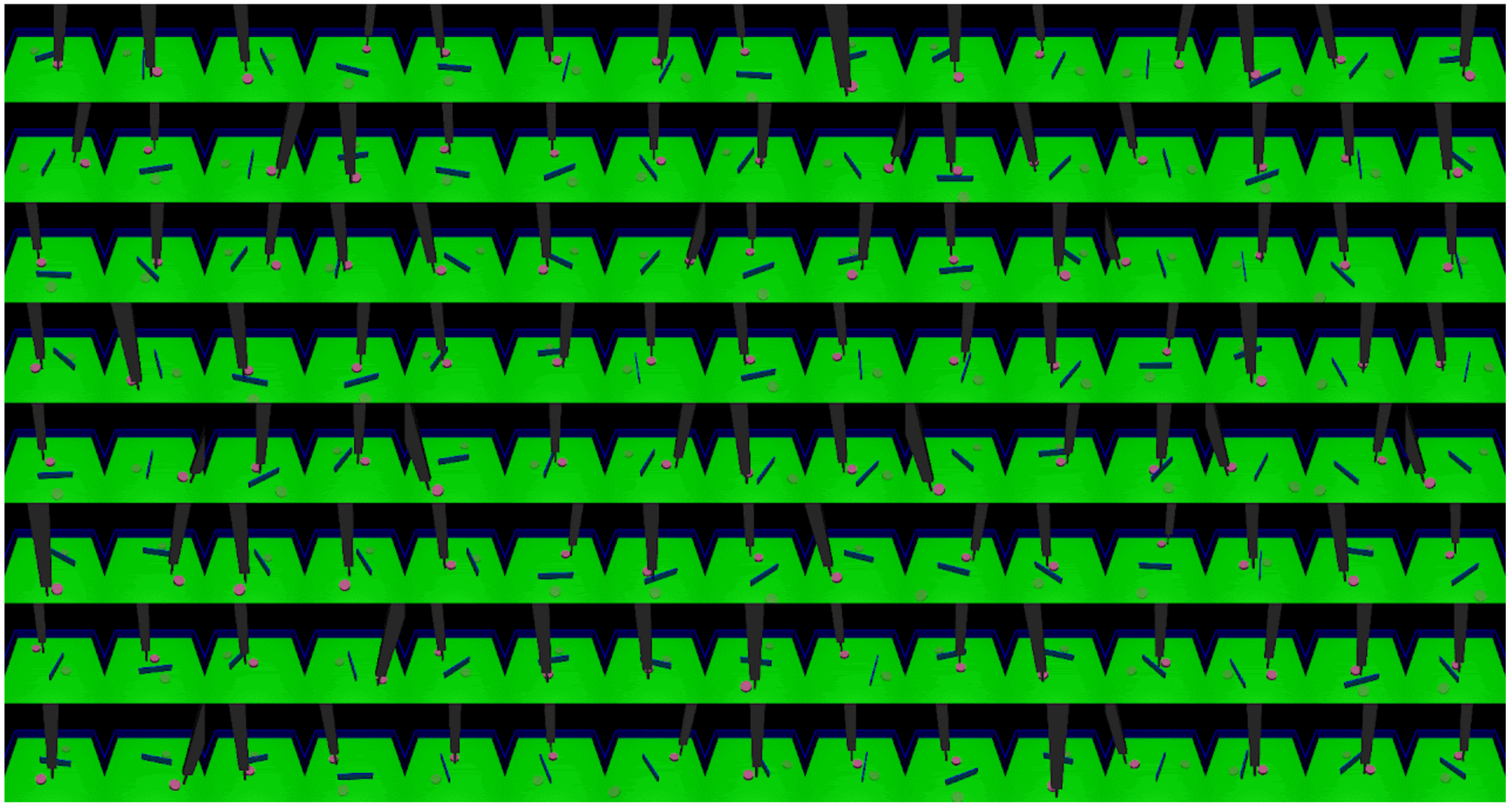


KeyIn — Keyframe Discovery for Visual Prediction and Planning



K. Pertsch, O. Rybkin, **J. Yang**, S. Zhou, K. Derpanis, J. Lim, K. Daniilidis, A. Jaegle. KeyIn: Keyframing for Visual Planning, Conference on Learning for Dynamics and Control, 2020.

KeyIn — Overview of Results



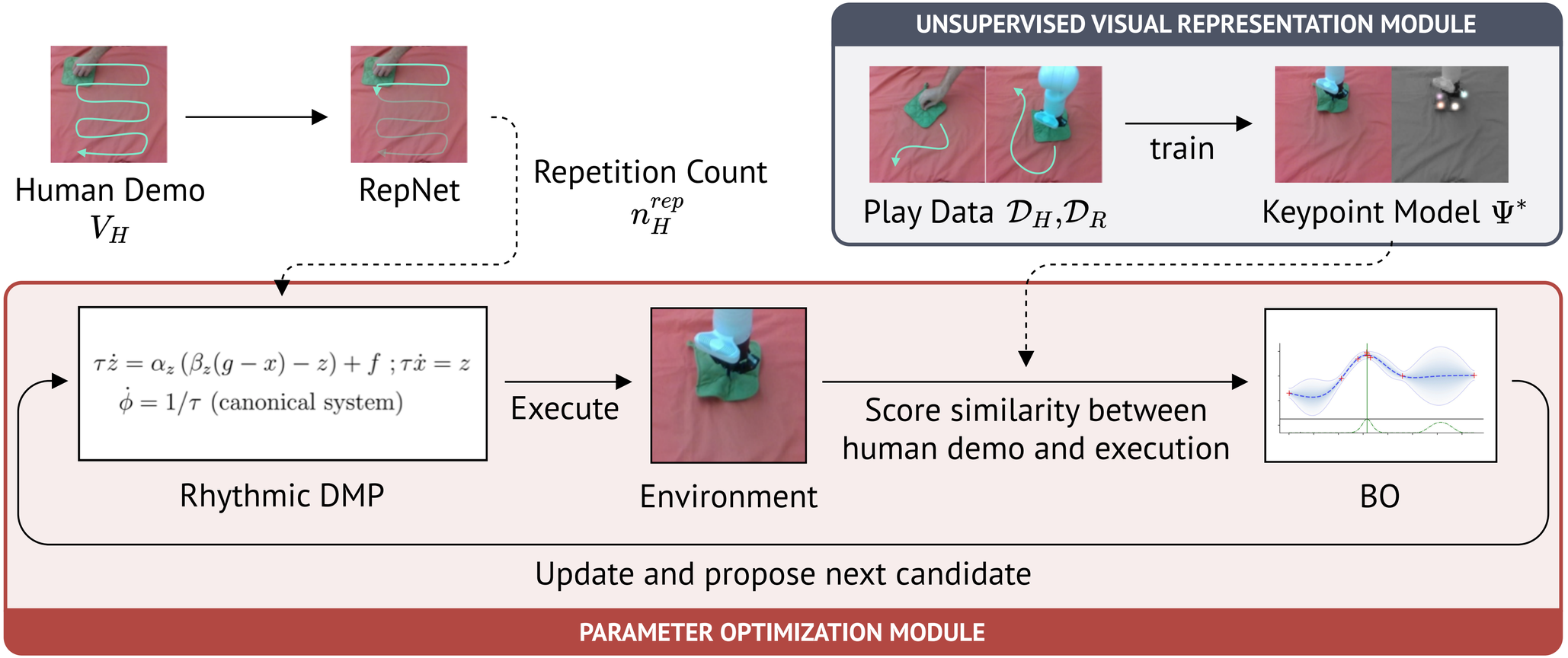
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Consider periodic manipulation tasks:



How can robots appropriately represent and learn periodic policies by watching humans?

ViPTL — A Framework for Learning Periodic Tasks

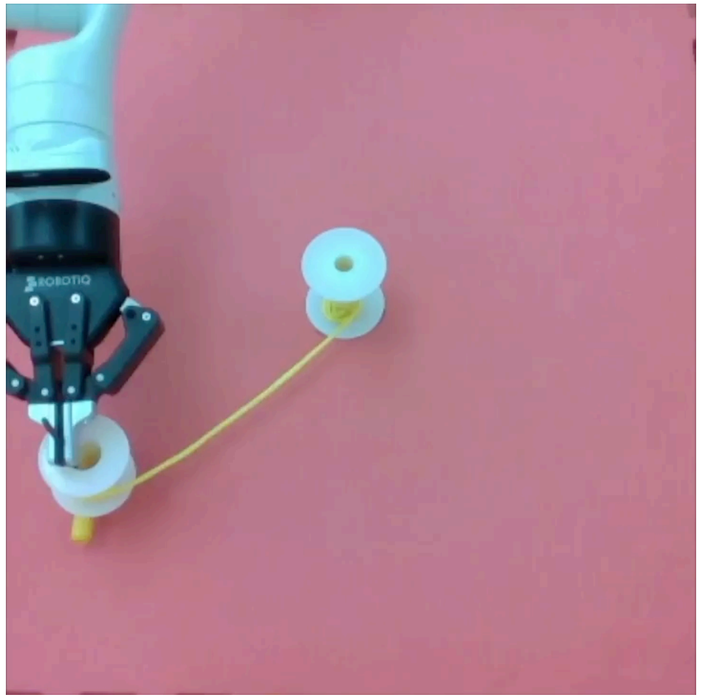


J. Yang, J. Zhang, C. Settle, A. Rai, R. Antonova, J. Bohg. Learning Periodic Tasks from Human Demonstrations, ArXiv, 2021.

ViPTL — Overview of Results



Table Wiping



Rope Winding



Food Stirring

J. Yang, J. Zhang, C. Settle, A. Rai, R. Antonova, J. Bohg. Learning Periodic Tasks from Human Demonstrations, ArXiv, 2021.

Agenda

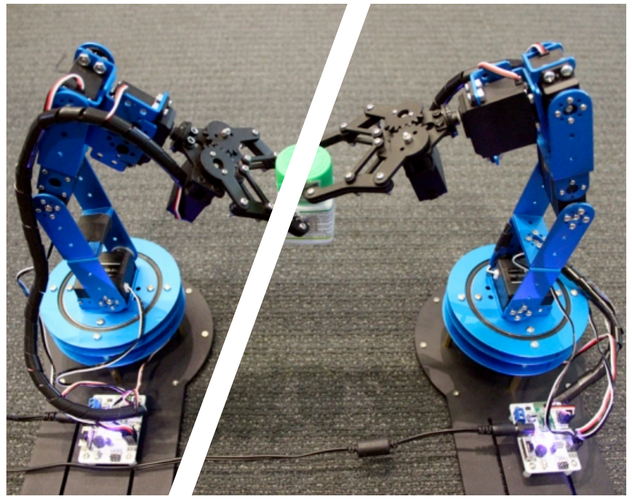
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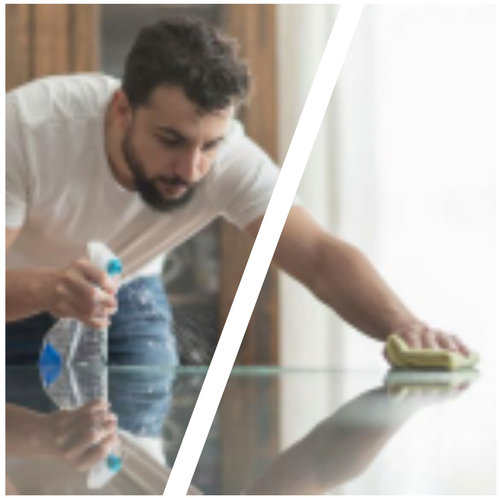
Consider multi-agent collaboration:



mix + hold



rotate + hold



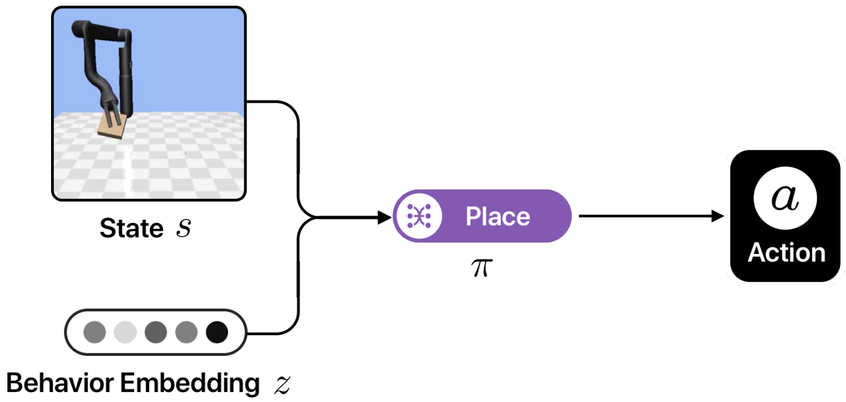
spray + mob

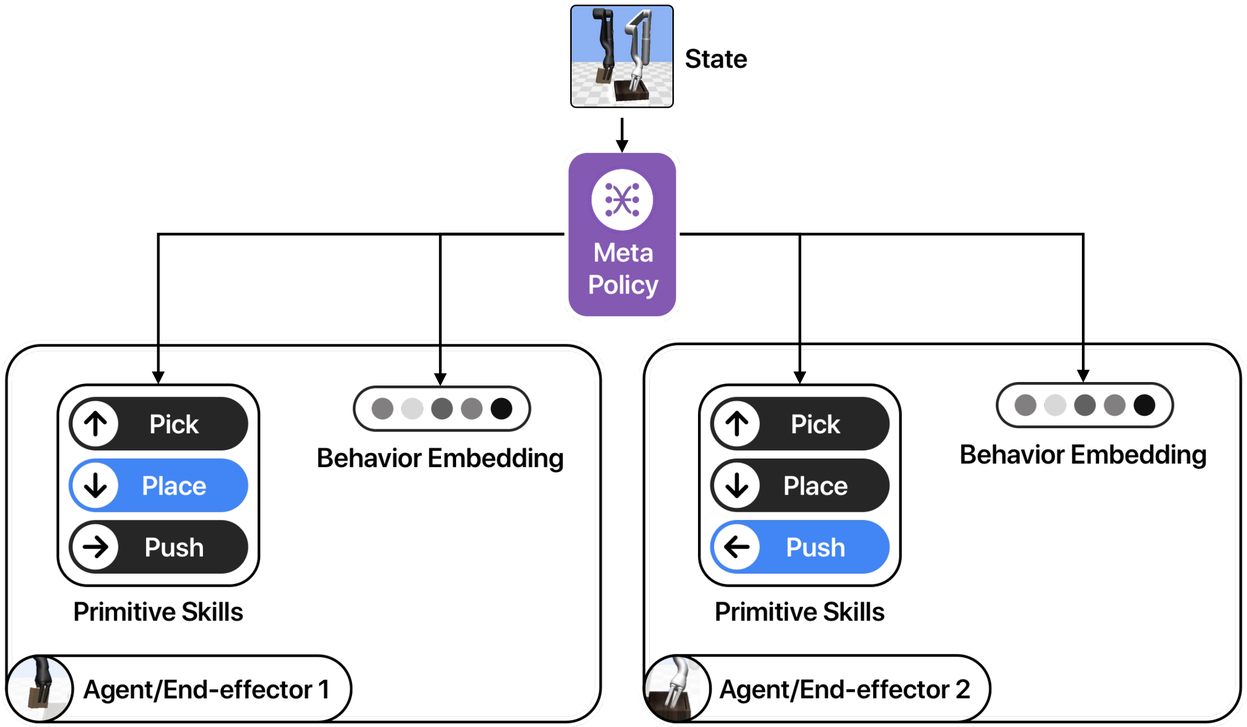


walk + control

Humans first learn to solve part of tasks.
Then, learn to **coordinate** multiple skills.

Learning to Coordinate Manipulation Skills



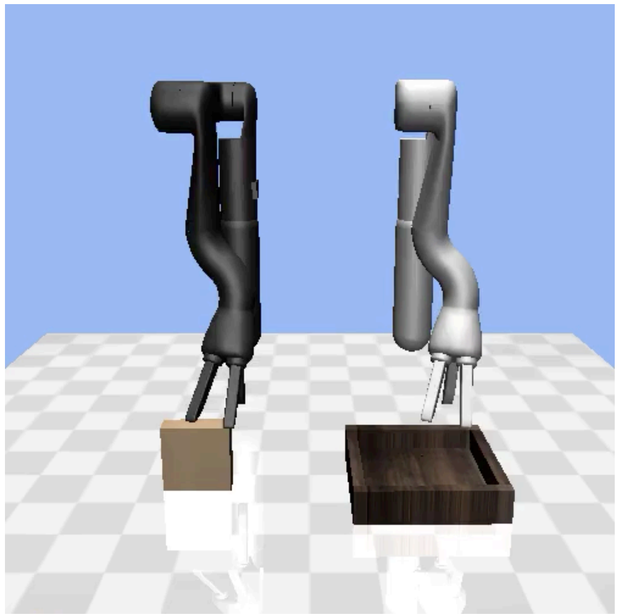


Phase 1: train primitive skills that support selection of diverse behaviors.

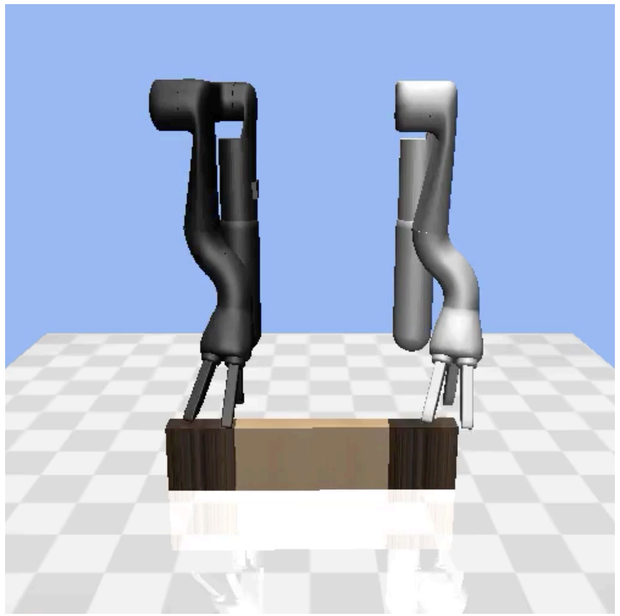
Phase 2: train a meta-policy to select skills and skill behaviors for each end-effector/agent.

Y. Lee, **J. Yang**, J. J. Lim. Learning to Coordinate Manipulation Skills via Skill Behavior Diversification, International Conference on Learning Representations, 2020.

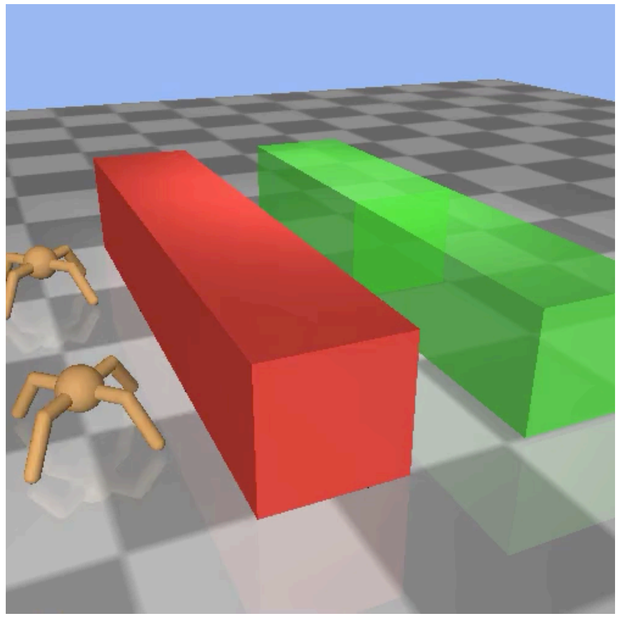
Overview of Results



Pick-Push-Place



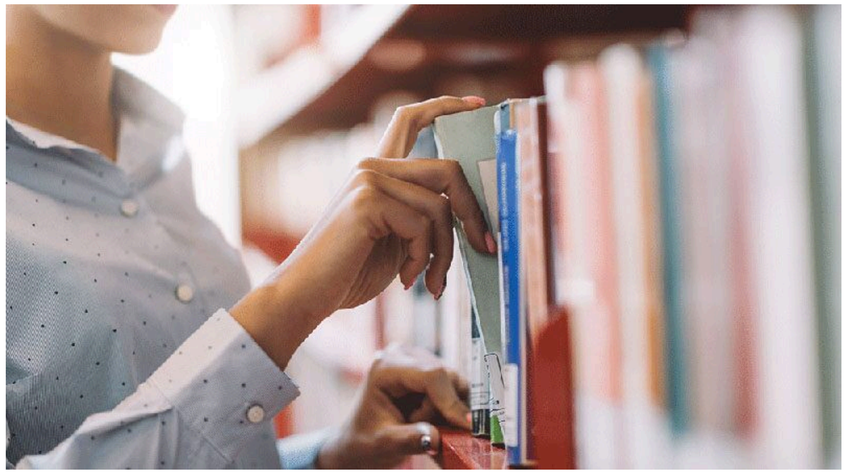
Bar moving



Ant push

Y. Lee, **J. Yang**, J. J. Lim. Learning to Coordinate Manipulation Skills via Skill Behavior Diversification, International Conference on Learning Representations, 2020.

Consider working with diverse objects across varying views:







Problems with Existing Works



- Easy to train
- No info about object appearance

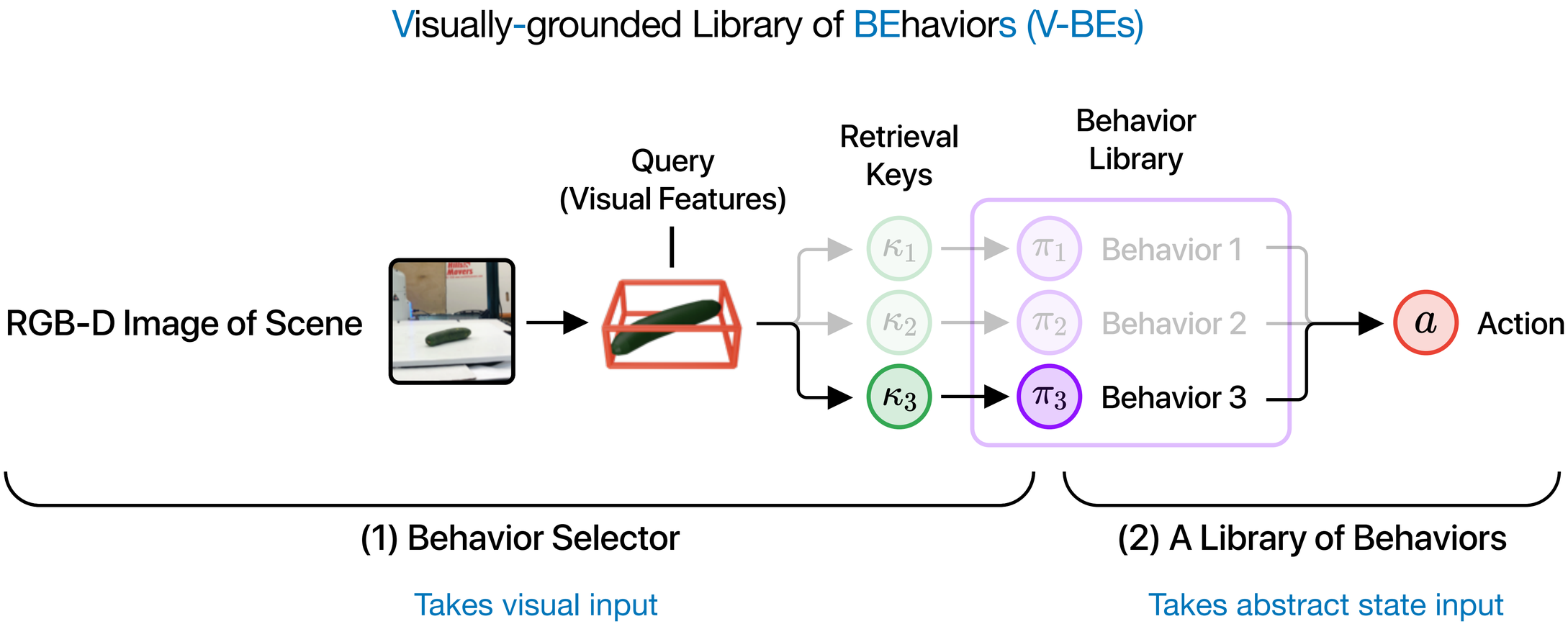


- Flexible
- Data aggressive
- Doesn't easily generalize across objects, views, and configurations

How can we combine the benefit of both sides?

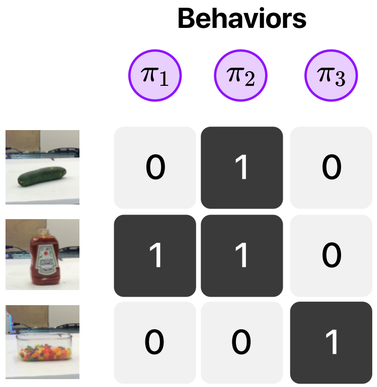
- Use abstract representations for efficient behavior learning
- Utilize semantically-rich representations for handling diverse objects and views

Overview of Proposed Framework

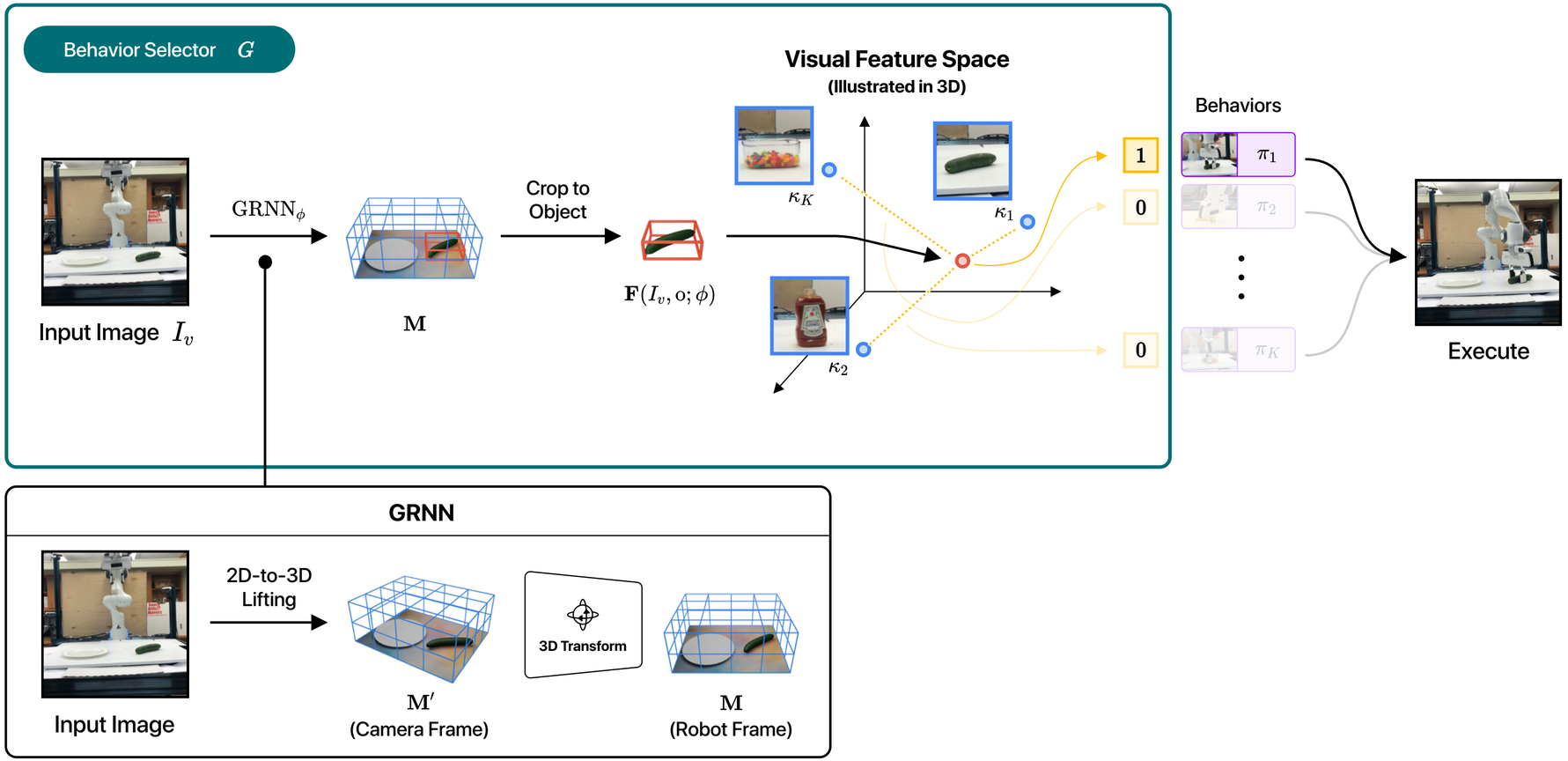


J. Yang*, H.-Y. Tung*, Y. Zhang*, G. Pathak, A. Pokle, C.G. Atkeson, K. Fragkiadaki. Visually-Grounded Library of Behaviors for Manipulating Diverse Objects across Diverse Configurations and Views, 5th Annual Conference on Robot Learning, 2021.

V-BEs: Visually-grounded Library of Behaviors

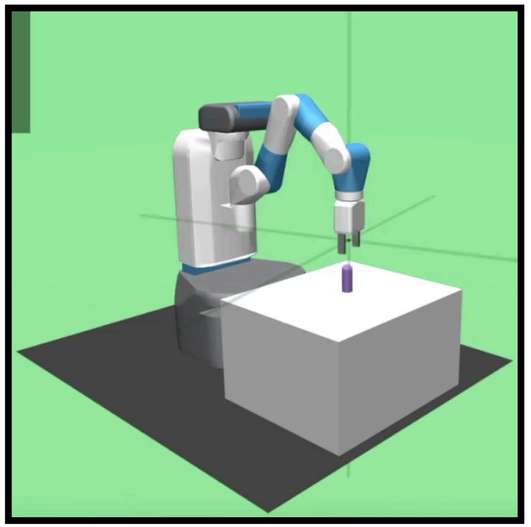


Robot interacts with training objects using different behaviors to self-collect binary success/failure affordance labels.

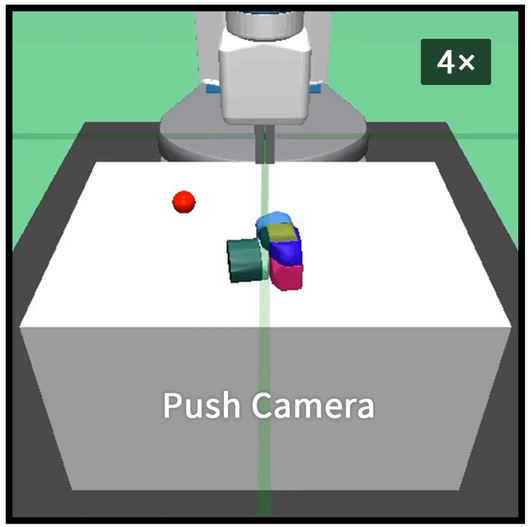


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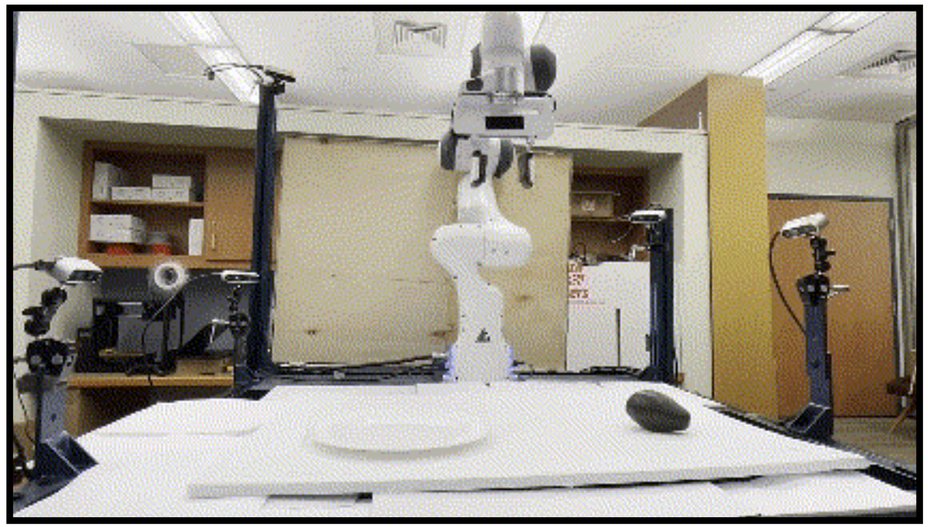
V-BEs: Overview of Results



Simulated Grasping



Simulated Pushing

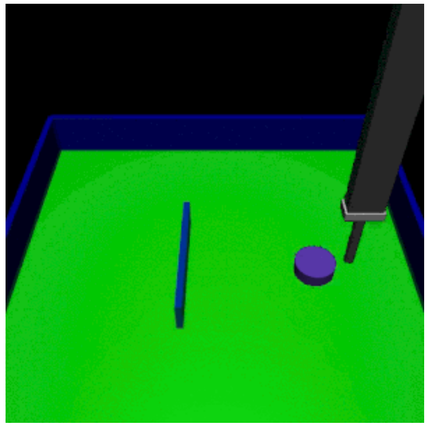


Real Robot Object Transporting

J. Yang*, H.-Y. Tung*, Y. Zhang*, G. Pathak, A. Pokle, C.G. Atkeson, K. Fragkiadaki. Visually-Grounded Library of Behaviors for Manipulating Diverse Objects across Diverse Configurations and Views, 5th Annual Conference on Robot Learning, 2021.

Summary

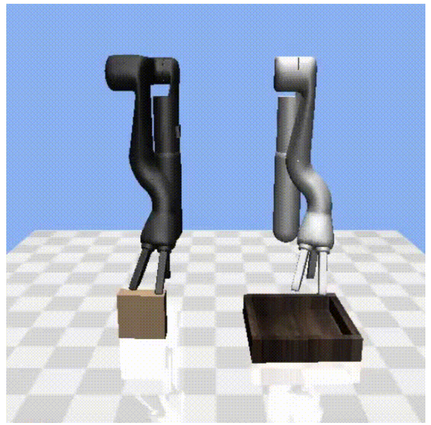
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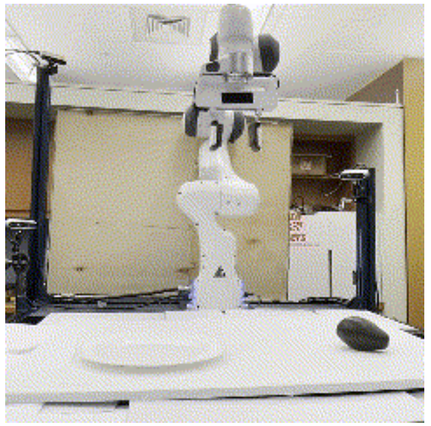
Keyframe-based Visual Planning



Representing Periodic Policies



Multi-agent Collaboration



Framework for Diverse Manipulation

Thank you!